



OBJECT ORIENTED SOFTWARE ENGINEERING

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How to generate a Class Model?

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- A class model captures the **static structure** of the system by characterizing
 - the **objects** in the system,
 - the **relationships between the objects**, and
 - the **attributes** and
 - **operations** for each class of objects

Object:

- Concept, abstraction, or a thing **with identity**
- **Proper nouns** or **specific reference** in problem description.

Class:

- A Class is a group of objects with
 - Same properties (attributes)
 - Behaviour (operations)
 - Kind of relationships

Class modelling:

- It describe the static data of objects and their relationships to one another.

Class diagrams

Graphic notation for **modelling classes** and their relationships.

Object diagrams

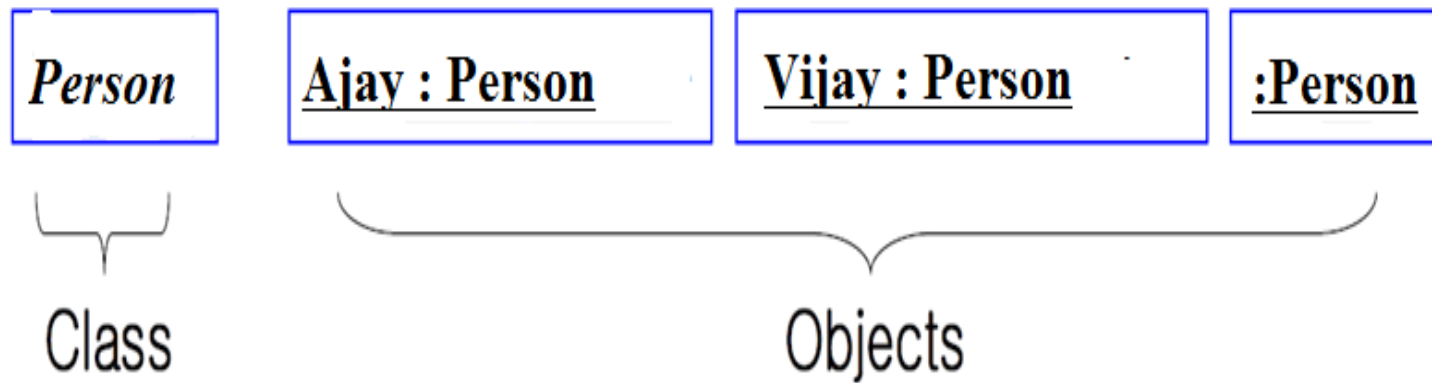
Graphic notation for **individual objects** and their relationships

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Object and Class Diagrams



For Example:



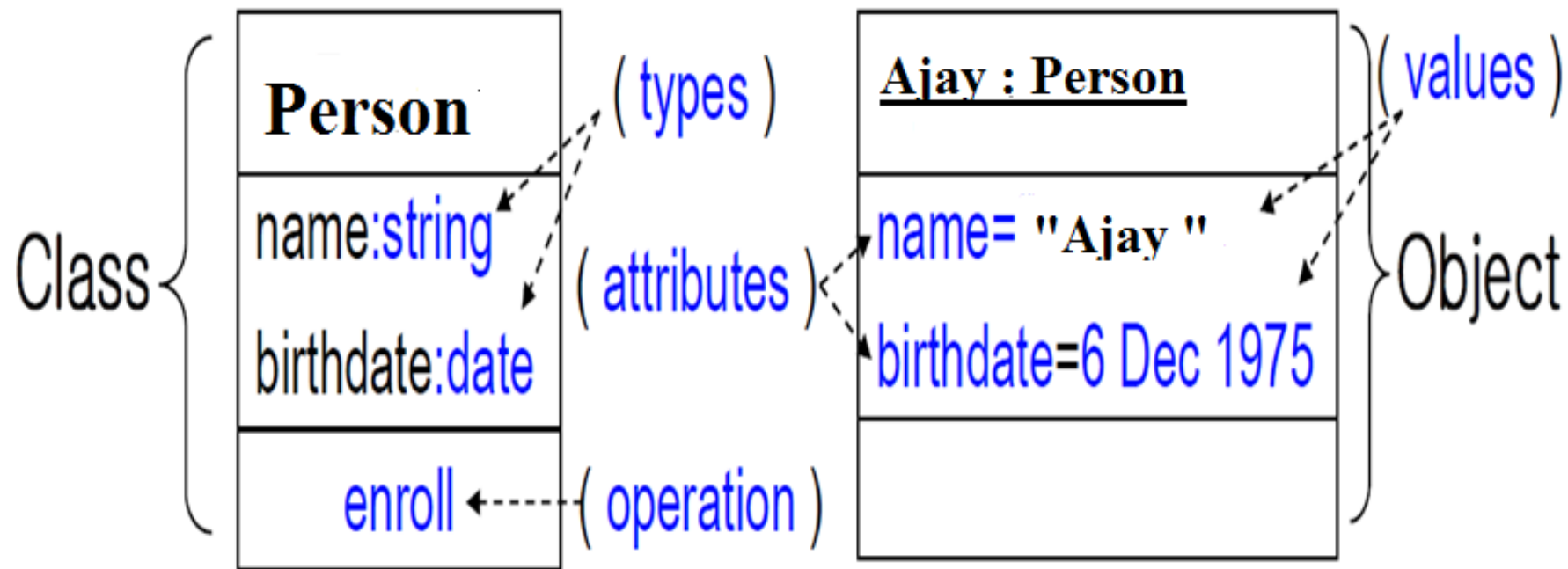
Note: classes & objects are the focus of class modeling

- **Value** : is a piece of data
- **Attribute** : value for each object. It is a named property of class.
- **Operations(Methods)**
Function or procedure that may be applied to or by objects in a class
- All objects in a class share the same operations

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Class and Objects with attributes

For Example:



- A box represents a class and may have as many as three compartments. It contains from top to bottom: class name, list of attributes, and list of operations.
- The **attribute** and **operation** compartments are **optional** and missing attribute compartment means that are unspecified.

ClassName
attributeName1 : dataType1 = defaultValue1 attributeName2 : dataType2 = defaultValue2 ...
operationName1 (argumentList1) : resultType1 operationName2 (argumentList2) : resultType2 ...

- UML symbol for both **class** and **objects** is rectangular box.
- Objects are modeled using box with object name followed by colon followed by class name and underline .
- Use **boldface** to list class name, **center** the name in box and **capitalize** first letter of name.

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Link and Association

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Links and Associations



- Links and Associations are the **means for establishing relationships** among objects and classes.

Then,

- What is a Link ?
- What is an Association?

Link:

- It is a physical or conceptual connection among objects.
- For Eg: Ajay *WorksFor* IBM Company.
- Most links relate two **objects**, but some links relate 3 or **more objects**.

Link:

- A link is **instance of an association**.
- UML notation for a link is a **line between objects**.
- It provides a relationship among objects.

Association:

- Group of links with **common structure** and **common semantics**
- It is **set of potential links** in the same way that a class describes a set of potential objects.
- Association is bi-directional

Association example:

- A Person works for company and inverse company employs person

association

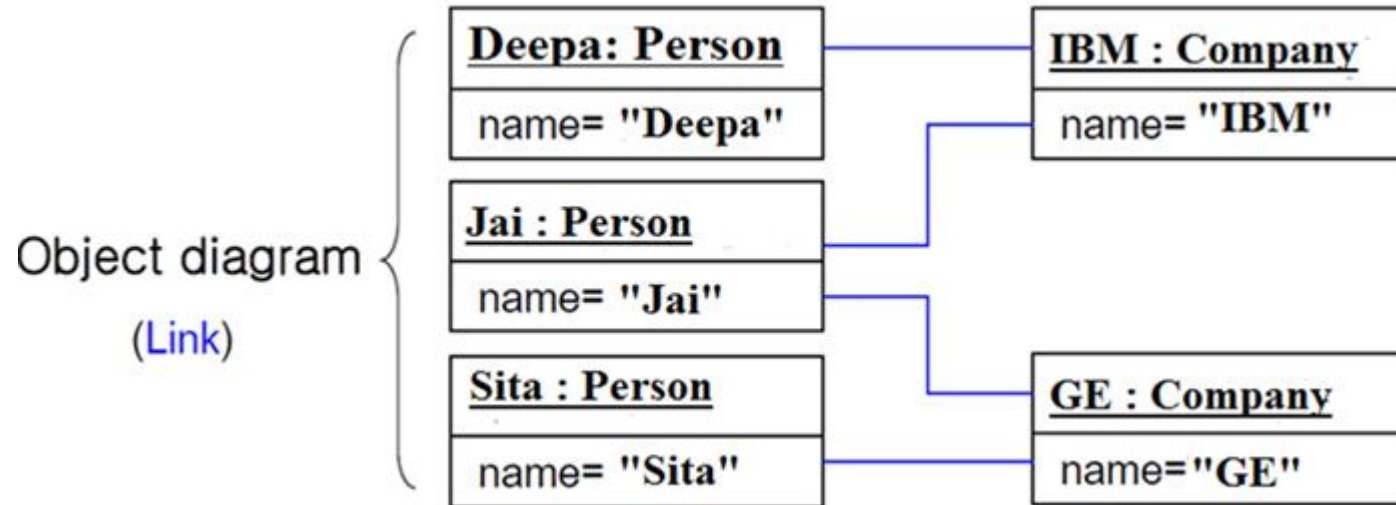
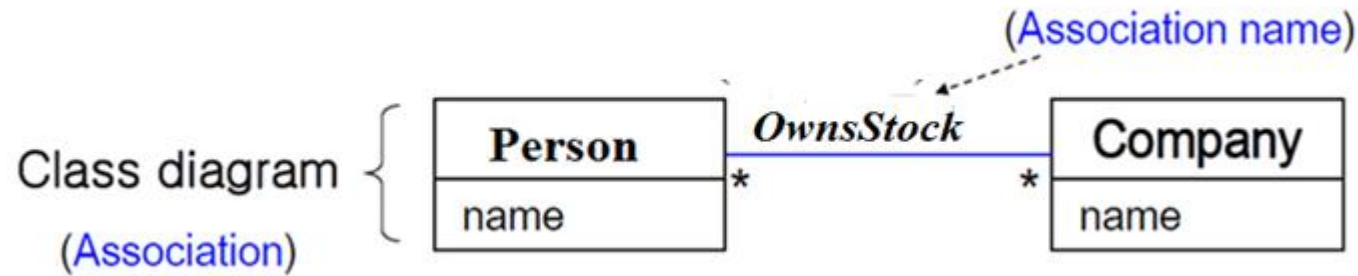


if association name is replaced with "owns>",
it would read "Class 1 owns Class 2"

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Links and Associations

Another example:



- Multiplicity constrains the number of related objects.
- It specifies the **number of instances of one class that may relate to a single instance of an associated class**.
- Assign a multiplicity to an **association end**

UML diagrams specifies the multiplicity with an interval such as

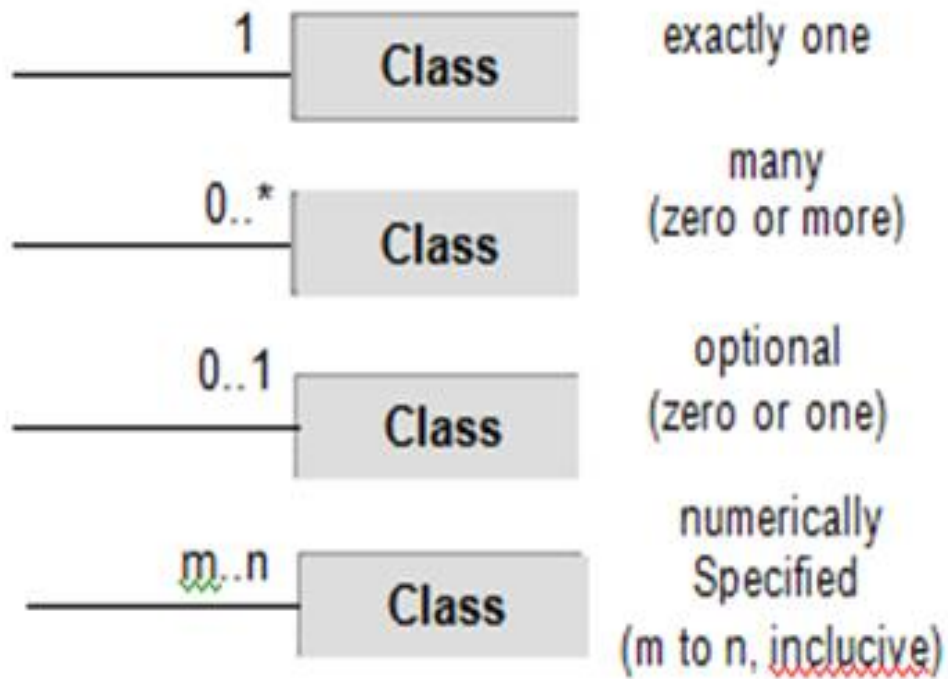
“1” → exactly one

“1..*” → one or more

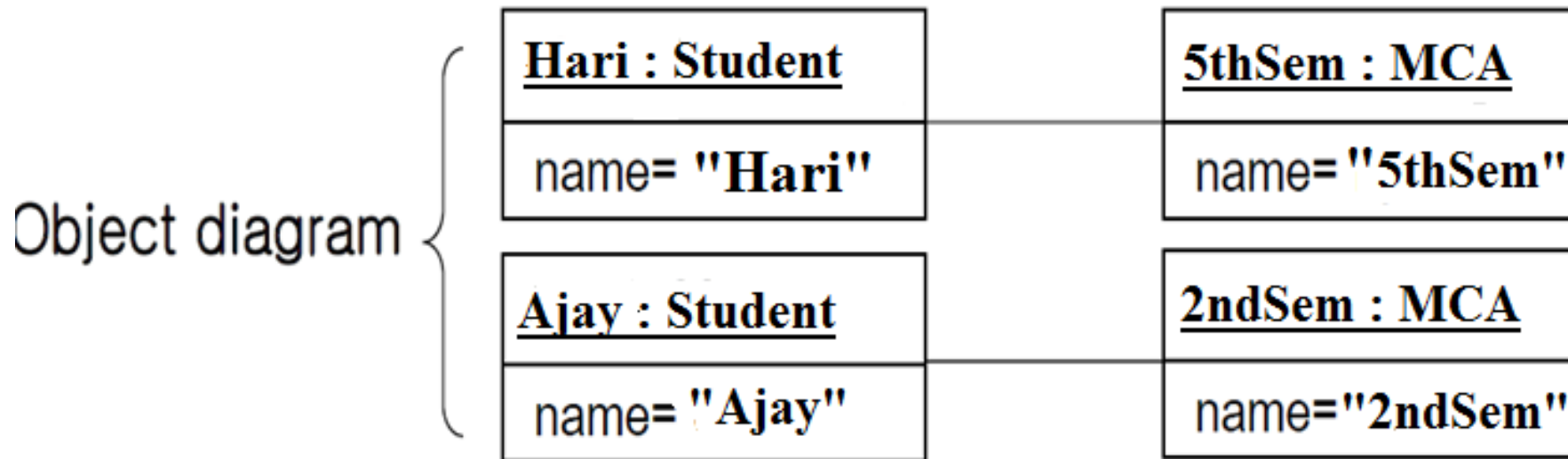
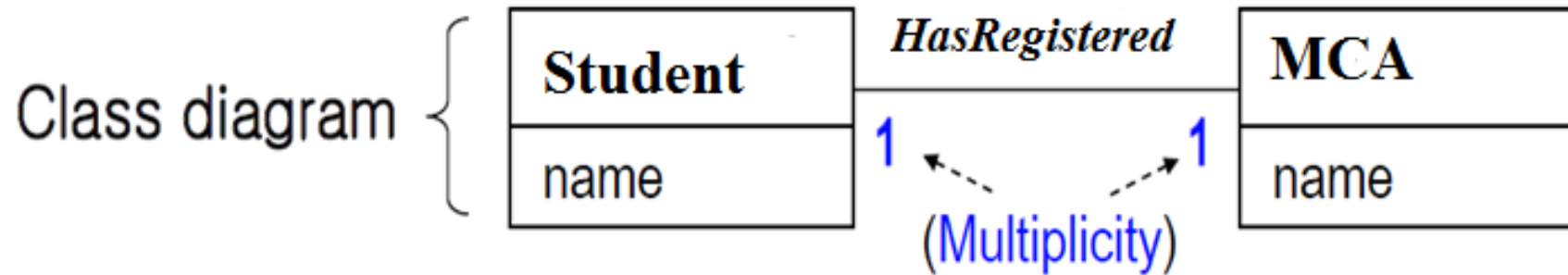
“3.5” → three to five (inclusive)

“*. * ” → many (i.e, zero to more)

Multiplicity notations in UML diagram

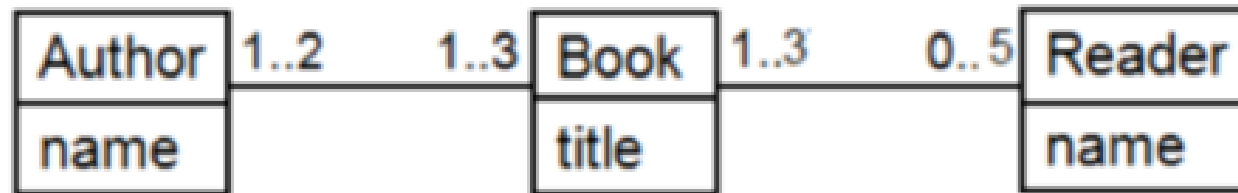


For Example:



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Activity1 : on Multiplicity

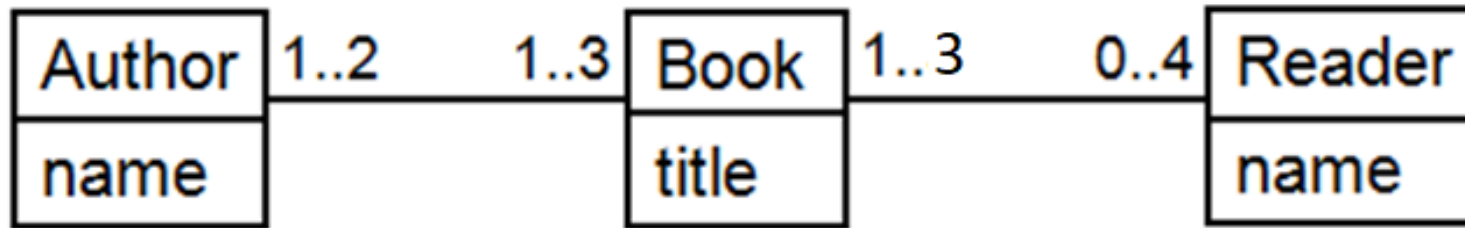


If there are 4 authors, then

- What is the minimum and maximum number of books?
What's the minimum and maximum number of readers?

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Activity 2: on Multiplicity

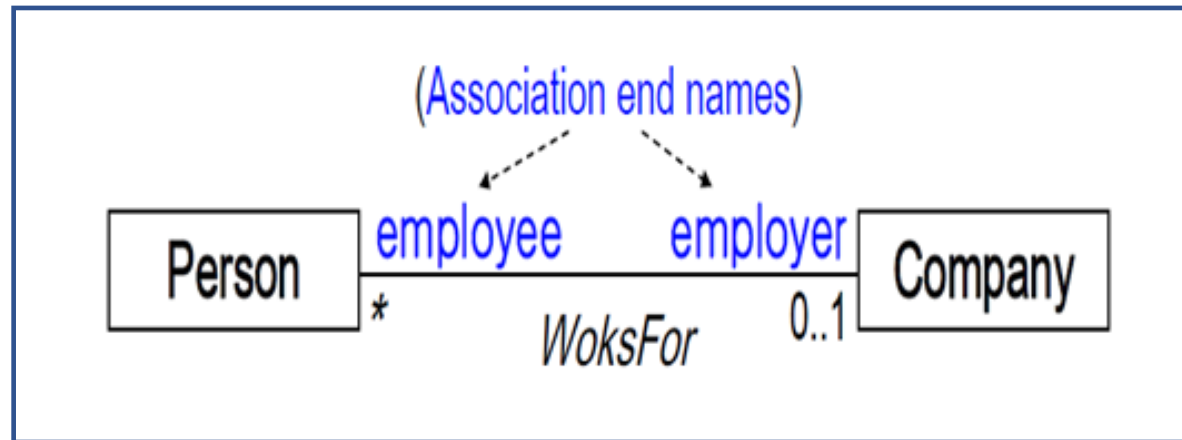


If there are 6 authors, then

- What is the minimum and maximum number of books?
- What is the minimum and maximum number of readers?
- If there are 7 authors then
- What is min and max number of books?
- What is min and max number of readers?

- Multiplicity implicitly refers to the end of association.
- Association ends can not only be assigned multiplicity, they **can also be named**.
- Association end names often appeared as **noun** in problem descriptions.
Each end of an association can have a name.

Association end names are convenience for **traversing** associations.



employee

employer

Ajay

IBM

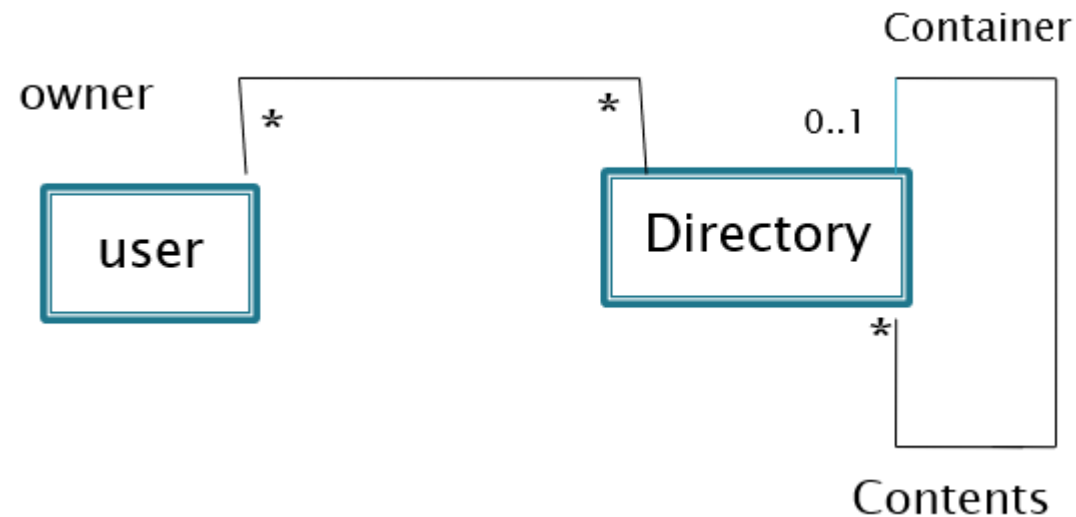
Vijay

GE

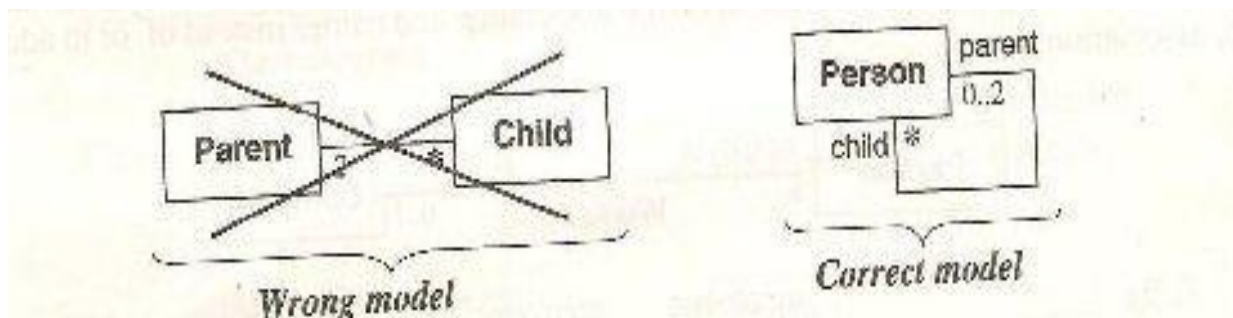
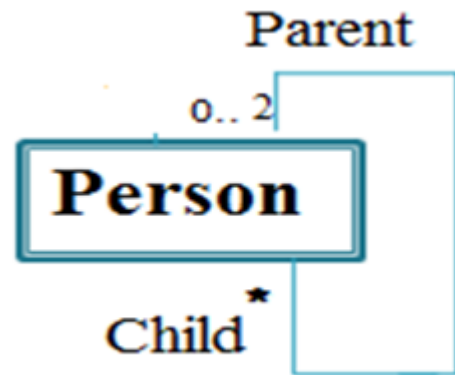
Sita

GE

- Association end names are necessary for **associations between two objects of the same class**(self association) they can also distinguish **multiple associations** between a pair of classes.



- Use association end names to model **multiple references to the same class**.





THANK YOU

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